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1. The relation R is not an equivalence relation. Equivalence relations are reflexive, symmetric, and transitive. However, R is not transitive, as shown below:

Reflexive: We know it's reflexive because xRx if and only if $|x - x| < 1$, which we know is true.

Irreflexive: It's not irreflexive because xRx is always true (since $|x - x| < 1$).

Symmetric: We know it's symmetric because if $|x - y| < 1$, then $|y - x| < 1$, since $|x - y| = |y - x|$.

Antisymmetric: It's not antisymmetric because xRy always implies yRx, regardless of whether or not $x = y$.

Asymmetric: This is the inverse of symmetry, so we know it is not asymmetric.

Transitive: It's not transitive, because if $|x - y| < 1$ and $|y - z| < 1$, we don't necessarily know that $|x - z| < 1$.

Trichotomous: It's not trichotomous. We know this because any values of x and y such that $x - y \geq 1$ would be counterexamples. $\square$

2. Let's first consider $xRy \iff x - y \in \mathbb{Z}$. This is reflexive because $x - x$ is an integer, it is symmetric because if $x - y$ is an integer then $-(x - y) = y - x$ is also an integer, and it is transitive because if $x - y$ and $y - z$ are integers, then $(x - y) + (y - z) = x - z$ is also an integer. Therefore, R is an equivalence relation.

Now, let's consider $xRy \iff x - y \in \mathbb{Q}$. This is reflexive because $x - x$ is rational, it is symmetric because if $x - y$ is rational then $-(x - y) = y - x$ is also rational, and it is transitive because if $x - y$ and $y - z$ are rational, then $x - z$ is also rational. Therefore, R is an equivalence relation. $\square$

3. That statement assumes that for each a there exists b such that aRb is true. However, this is not always the case, so we can't assume that reflexivity follows from symmetry and transitivity. For example, consider the set $S = \{1, 4, 5\}$ and the relation R, such that $xRy \iff x + y > 3$. In this case, xRy is always true, and yRx is always true. However, xRx does not always hold, because $1 + 1 \not> 3$. $\square$

4. The first relation is $\sim$ on X where X is the set of all triangles in the plane, and we say that $x \sim y$ if triangles x and y are similar. We can see that $\sim$ is reflexive because all triangles are similar to themselves, $\sim$ is symmetric because if triangle x is similar to triangle y, then y is similar to x, and $\sim$ is transitive because if x and y are similar and y and z are similar, then x and z are similar. So $\sim$ is an equivalence relation.

The second relation is $\sim$ where $f : X \to Y$ is a function on X and $x_1 \sim x_2 \iff f(x_1) = f(x_2)$. We know that $\sim$ is reflexive because $f(x_1) = f(x_1)$, $\sim$ is symmetric because if $f(x_1) = f(x_2)$ then $f(x_2) = f(x_1)$, and $\sim$ is symmetric because if $f(x_1) = f(x_2)$ and $f(x_2) = f(x_3)$ then $f(x_1) = f(x_3)$. Therefore, $\sim$ is an equivalence relation. $\square$

5. The relation R is not guaranteed to be transitive, even if $\sim$ and $\equiv$ are.

For a counterexample, let's consider the set $S = \{1, 2, 3, 4\}$. Let's say
$\sim = \{(1,1), (2,2), (3,3), (4,4), (1,2), (2,1), (1,3), (3,1), (2,3), (3,2)\}$, where each pair in that set represents an element of S being paired by the relation $\sim$ to another element in S. We see that $\sim$ is reflexive because it includes the pairs $(1, 1)$, $(2, 2)$, $(3, 3)$, and $(4, 4)$. It is also symmetric because, for every pair (a, b) in $\sim$, there also exists another pair (b, a). We can also see that $\sim$ is transitive by checking that for each pair of pairs, (a, b) and (c, d), there exists another pair (a, d) in $\sim$. Therefore, $\sim$ is an equivalence relation.

Now, let's say $\equiv = \{(1,1), (2,2), (3,3), (4,4), (3,4), (4,3)\}$. By the same logic as before, we can show that $\equiv$ is an equivalence relation.

Now, we know that $R = \sim \cup \equiv$, which is $\{(1,1), (2,2), (3,3), (4,4), (1,2), (2,1), (1,3), (3,1), (2,3), (3,2), (3,4), (4,3)\}$. We can see immediately that R is not transitive, because $2R3$ and $3R4$ hold true, but $2R4$ does not. That means R is not an equivalence relation. This counterexample shows that even if $\sim$ and $\equiv$ are equivalence relations, R is not necessarily an equivalence relation. $\square$

6. We know that R is reflexive because if xRx, then either $x \sim x$ or $x \equiv x$, and since both $\sim$ and $\equiv$ are symmetric, xRx must be too.

Also, R is symmetric because if xRy, then $x \sim y$ and $x \equiv y$, and since those two are symmetric, $y \sim x$ and $y \equiv x$, so yRx as well.

It is also transitive, because if xRy and yRz, then $x \sim y$, $y \sim z$, $x \equiv y$, and $y \equiv z$ are all true. Since $\sim$ and $\equiv$ are transitive, $x \sim z$ and $x \equiv z$, which means xRz as well. $\square$

7. We know that no two equivalence classes overlap, because then they would be the same one. That means, since there are n elements and each equivalence class has k elements, the greatest number of equivalence classes possible is $\frac{n}{k}$. However, since $\sim$ is an equivalence relation, it is reflexive for all elements in X. That means every element is part of an equivalence class, so the equivalence classes cover all elements of X. This means the number of equivalence classes cannot be less than $\frac{n}{k}$. Therefore, there are $\frac{n}{k}$ equivalence classes. $\square$

8. I didn't really understand the definition of $\sim$ for this one.

9. We can see that $\sim$ is reflexive because $ad - bc = ab - ba = 0$. It is also symmetric because if $ad = bc$, then $cb = da$. Finally, it is transitive because, given the pairs $(a_1, a_2), (b_1, b_2), (c_1, c_2)$, if $a_1 b_2 = a_2 b_1$ and $b_1 c_2 = b_2 c_1$, we can multiply both equations to get $a_1 b_2 b_1 c_2 = a_2 b_1 b_2 c_1$, or $a_1 c_2 = a_2 c_1$. $\square$